WORKED SOLUTION

STEP I 1996, Question 9

Mechanics • Elastic Strings and Energy

STEP I 1996, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q9

The key idea

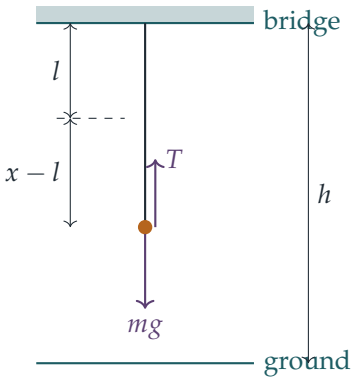
Everything here is energy. Gravity and the rope's tension are the only forces doing work, and both are conservative, so we never need the equation of motion in full. "Only just avoids the ground" means her speed is zero exactly when she reaches the ground. The maximum speed comes where the acceleration changes sign, which is where the tension balances her weight.

Solution

We model the jumper as a particle starting from rest at the bridge, the rope as light and obeying Hooke's law, and we ignore air resistance. Let x be the distance she has fallen. For $x \leq l$ the rope is slack and she is in free fall. For $x > l$ the extension is $x - l$, and Hooke's law gives tension

$$T = \frac{mg}{k} \cdot \frac{x - l}{l},$$

with elastic energy $\dfrac{mg}{k} \cdot \dfrac{(x-l)^2}{2l}$ stored in the rope.



The height of the bridge

At the lowest point she is momentarily at rest. She only just avoids the ground, so this lowest point is the ground itself, at $x = h$. Clearly $h > l$: if the rope never went taut she would be in free fall and would hit the ground at speed. Between leaving the bridge and the lowest point the kinetic energy starts and ends at zero, so the gravitational energy lost all goes into the rope:

$$mgh = \frac{mg}{k} \cdot \frac{(h-l)^2}{2l}.$$

Multiplying by $2kl/(mg)$ gives $2klh = (h-l)^2 = h^2 - 2hl + l^2$, so

$$h^2 - 2hl(k+1) + l^2 = 0,$$

as required. By the quadratic formula,

$$h = l(k+1) \pm l\sqrt{(k+1)^2 - 1} = l\left(k + 1 \pm \sqrt{k^2 + 2k}\right).$$

Ask yourself Which root do we want, and why does the other one appear at all?

Since $k > 0$ we have $\sqrt{k^2+2k} > \sqrt{k^2} = k$, so the minus sign gives $h < l(k+1-k) = l$. But the energy equation assumed the rope was stretched by $h - l$, which only makes sense for $h > l$. The spurious root comes from squaring: it gives the same value of $(h-l)^2$ for a "compression" the rope can't actually provide. (The two roots multiply to l^2, so one is above l and the other below.) Hence

$$h = l\left(k + 1 + \sqrt{k^2 + 2k}\right).$$

The maximum speed

While the rope is slack she accelerates downwards at g. Once it is taut, Newton's second law downwards gives

$$m\ddot{x} = mg - \frac{mg}{k} \cdot \frac{x-l}{l}.$$

This is positive while $x - l < kl$ and negative after that. So she speeds up until the extension is kl, the point where the tension equals her weight, and slows down from then on. Her maximum speed is therefore reached at $x = l + kl$. By conservation of energy from the bridge to this point,

$$\begin{aligned} \tfrac{1}{2}mv^2 &= mg(l+kl) - \frac{mg}{k} \cdot \frac{(kl)^2}{2l} \\ &= mgl(1+k) - \tfrac{1}{2}mgkl. \end{aligned}$$

Doubling and dividing by m,

$$v^2 = 2gl + 2kgl - kgl = (k+2)gl,$$

as required.

Checks

- As $k \to 0$ the rope becomes very stiff. Then $h \to l$ (she barely stretches it), and $v^2 \to 2gl$, the free-fall speed after a drop of l, which is right since the rope stops her almost at once.
- Stepping the equation of motion numerically for random k and l gives a lowest point and maximum speed agreeing with the formulae to five decimal places. For example, $k = 1.5$, $l = 10\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ gives $h = 10(2.5 + \sqrt{5.25}) \approx 47.9\,\mathrm{m}$ and $v^2 = 343$, so $v \approx 18.5\,\mathrm{m\,s^{-1}}$.

Common slips

- Taking the maximum speed at the moment the rope goes taut. She is still accelerating then, because the tension is zero.
- Using the full drop x instead of the extension $x - l$ in the elastic energy.
- Keeping both roots for h without checking that $h > l$.

Beyond STEP

Once the rope is taut the motion is simple harmonic about the equilibrium extension kl, so the second half of the fall is part of a sine curve with angular frequency $\sqrt{g/(kl)}$. Piecewise systems like this, which switch between free fall and an oscillator, are a nice introduction to how energy methods and phase-plane pictures handle non-smooth forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy conservation and oscillations carefully at this level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.